

8 PHASES OF THE CYBER KILL CHAIN

1

Reconnaissance

2

Intrusion

3

Exploitation

4

Privilege Escalation

5

Lateral Movement

6

Obfuscation / Anti-forensics

7

Denial of Service

8

Exfiltration

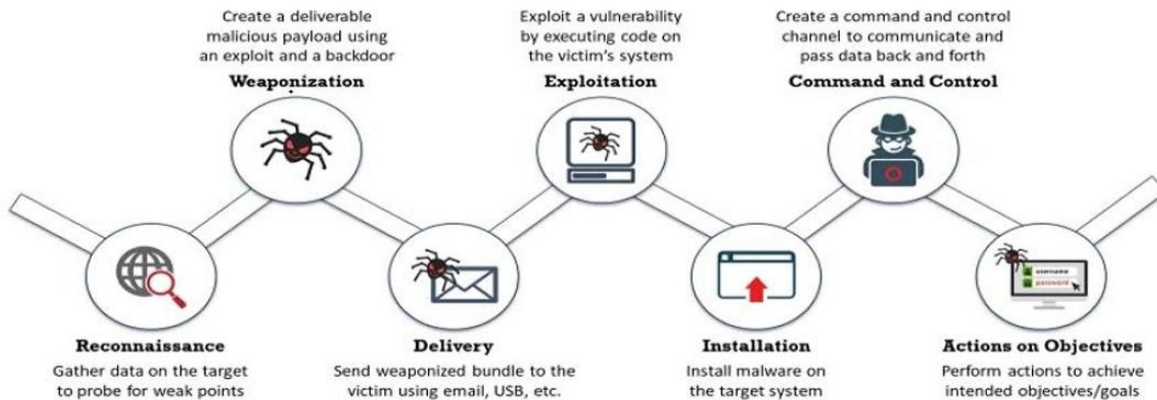
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Cyber Kill Chain Methodology:

The cyber kill chain is essentially a cyber-security model created by “Lockheed Martin” that traces the stages of a cyberattack, identifies vulnerabilities, and helps security teams to stop the attacks at every stage of the chain

cyber kill chain methodology



Reconnaissance:-

An observation stage in which attacker collects data about the target and the tactics for the attack. This includes harvesting email addresses and gathering other information.

Detect:

Web Analytics; Threat Intelligence; Network Intrusion Detection System.

Deny:

Information Sharing Policy; Firewall Access Control Lists.

Weaponization:-

Preparing or creating a deliverable malicious payload by attacker using an exploit and a backdoor.

Detect:

Threat Intelligence; Network Intrusion Detection System.

Deny: Network Intrusion Prevention System.

Delivery:-

The attacker delivers the weaponized malware phishing email or some other medium. The most common delivery vectors for weaponized payloads include websites, removable disks, and emails. This is the most important stage where the attack can be stopped by the security teams.

Detect: Endpoint Malware Protection.

Deny: Change Management; Application Whitelisting; Proxy Filter; Host-Based Intrusion Prevention System, Router ACL, Trust Zone etc.

Exploit:-

When malicious code is delivered into the organization's system. Exploiting vulnerability by executing that malicious code on the victim's system. It means the perimeter is breached here.

Detect:

Endpoint Malware Protection; Host-Based Intrusion Detection System.

Deny:

Secure Password; Patch Management, Firewall, Trust Zones.

Installation:-

A backdoor or Remote Access Trojan is installed on the target system by the malware that provides access to the intruder. This is also another important stage where the attack can be stopped using systems such as HIPS (Host-based Intrusion Prevention System). **Detect:**

Security Information and Event Management (SIEM); Host-Based Intrusion Detection System

Deny:

Privilege Separation; Strong Passwords; Two-Factor Authentication, Router ACL, Firewall, Trust Zone.

Command & Control:-

The attacker gains control over the organization's systems and network. Attackers gain access to privileged accounts and attempt brute force attacks, search for credentials, and change permissions to take over the control. **Detect:**

Network Intrusion Detection System; Host-Based Intrusion Detection System

Deny:

Firewall Access Control Lists; Network Segmentation

Actions on Objective:-

The attacker finally extracts the data from the system. The objective involves gathering, encrypting, and extracting confidential information from the organization's environment.

Endpoint Malware Protection, Incident Response, Data Loss Prevention; Security Information and Event Management (SIEM).